

## SAFETY DATA SHEET

Creation Date 24-November-2010

Revision Date 27-March-2024

Revision Number 4

### 1. Identification

**Product Name** Cerium(IV) ammonium nitrate

**Cat No. :** A12882

**CAS-No** 16774-21-3  
**Synonyms** Ceric ammonium nitrate; CAN

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Oxidizing solids</b>                                 | Category 2   |
| <b>Corrosive to metals</b>                              | Category 1   |
| <b>Acute oral toxicity</b>                              | Category 4   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 C |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Skin Sensitization</b>                               | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

May intensify fire; oxidizer  
May be corrosive to metals  
Harmful if swallowed  
Causes severe skin burns and eye damage  
May cause an allergic skin reaction  
May cause respiratory irritation

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep/Store away from clothing/combustible materials  
Take any precaution to avoid mixing with combustibles  
Keep only in original container  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Absorb spillage to prevent material damage  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up  
Store in corrosive resistant polypropylene container with a resistant inliner  
Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                                               | CAS-No     | Weight % |
|---------------------------------------------------------|------------|----------|
| Cerate(2-), hexakis(nitrato-O)-, diammonium, (OC-6-11)- | 16774-21-3 | <100     |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.                                                                                                                                                                                                          |
| <b>Ingestion</b>                       | Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: May cause methemoglobinemia |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                          |
| <b>Flash Point</b>                      | No information available                                          |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | No information available                                          |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Oxidizing Properties</b>             | Oxidizer                                                          |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Ammonia. Heavy metal oxides. Thermal decomposition can lead to release of irritating gases and vapors.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
3

**Instability**  
3

**Physical hazards**  
OX

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.                                                                                                                                                              |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Do not allow material to contaminate ground water system. Should not be released into the environment. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.                                                     |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from clothing and other combustible materials. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Corrosives area. Incompatible Materials. Acids. Bases. Cyanides. Metals. Reducing Agent. Finely powdered metals. Strong reducing agents. Combustible material.        |

## 8. Exposure controls / personal protection

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Exposure Guidelines</u></b>  | This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies.                                                                                                                                                                                                                                                                                                              |
| <b><u>Engineering Measures</u></b> | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.<br>Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |

### **Personal protective equipment**

|                        |                   |
|------------------------|-------------------|
| <b>Eye Protection</b>  | Goggles           |
| <b>Hand Protection</b> | Protective gloves |

| <b>Glove material</b>                               | <b>Breakthrough time</b>          | <b>Glove thickness</b> | <b>Glove comments</b>  |
|-----------------------------------------------------|-----------------------------------|------------------------|------------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | -                      | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                 |
|----------------------------------------|---------------------------------|
| Physical State                         | Powder Solid                    |
| Appearance                             | Orange                          |
| Odor                                   | pungent                         |
| Odor Threshold                         | No information available        |
| pH                                     | 1 @ 20°C 50 g/l aq.sol          |
| Melting Point/Range                    | 107 - 108 °C / 224.6 - 226.4 °F |
| Boiling Point/Range                    | No information available        |
| Flash Point                            | No information available        |
| Evaporation Rate                       | Not applicable                  |
| Flammability (solid,gas)               | No information available        |
| Flammability or explosive limits       |                                 |
| Upper                                  | No data available               |
| Lower                                  | No data available               |
| Vapor Pressure                         | No information available        |
| Vapor Density                          | Not applicable                  |
| Specific Gravity                       | No information available        |
| Solubility                             | No information available        |
| Partition coefficient; n-octanol/water | No data available               |
| Autoignition Temperature               | No information available        |
| Decomposition Temperature              | No information available        |
| Viscosity                              | Not applicable                  |
| Molecular Formula                      | H8 Ce N8 O18                    |
| Molecular Weight                       | 548.22                          |

## 10. Stability and reactivity

|                                  |                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                          |
| Stability                        | Stable under normal conditions. Oxidizer: Contact with combustible/organic material may cause fire.                          |
| Conditions to Avoid              | Excess heat. Incompatible products. Combustible material.                                                                    |
| Incompatible Materials           | Acids, Bases, Cyanides, Metals, Reducing Agent, Finely powdered metals, Strong reducing agents, Combustible material         |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Ammonia, Heavy metal oxides, Thermal decomposition can lead to release of irritating gases and vapors |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                     |
| Hazardous Reactions              | None under normal processing.                                                                                                |

## 11. Toxicological information

**Acute Toxicity****Product Information****Component Information**

| Component                        | LD50 Oral            | LD50 Dermal         | LC50 Inhalation |
|----------------------------------|----------------------|---------------------|-----------------|
| Cerate(2-), hexakis(nitrato-O)-, | 300-2000 mg/kg (Rat) | >2000 mg/kg ( Rat ) | Not listed      |

|                        |  |  |  |
|------------------------|--|--|--|
| diammonium, (OC-6-11)- |  |  |  |
|------------------------|--|--|--|

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** Irritating to eyes, respiratory system and skin

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                               | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Cerate(2-), hexakis(nitrato-O)-, diammonium, (OC-6-11)- | 16774-21-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: May cause methemoglobinemia

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

**Persistence and Degradability** May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| UN-No                   | UN3085                                                |
| Proper Shipping Name    | consumer commodity Oxidizing solid, corrosive, n.o.s. |
| Technical Name          | Cerium(IV) ammonium nitrate                           |
| Hazard Class            | 5.1                                                   |
| Subsidiary Hazard Class | 8                                                     |
| Packing Group           | II                                                    |

### TDG

|                         |                                    |
|-------------------------|------------------------------------|
| UN-No                   | UN3085                             |
| Proper Shipping Name    | Oxidizing solid, corrosive, n.o.s. |
| Hazard Class            | 5.1                                |
| Subsidiary Hazard Class | 8                                  |
| Packing Group           | II                                 |

### IATA

|                         |                                    |
|-------------------------|------------------------------------|
| UN-No                   | UN3085                             |
| Proper Shipping Name    | Oxidizing solid, corrosive, n.o.s. |
| Hazard Class            | 5.1                                |
| Subsidiary Hazard Class | 8                                  |
| Packing Group           | II                                 |

### IMDG/IMO

|                         |                                    |
|-------------------------|------------------------------------|
| UN-No                   | UN3085                             |
| Proper Shipping Name    | Oxidizing solid, corrosive, n.o.s. |
| Hazard Class            | 5.1                                |
| Subsidiary Hazard Class | 8                                  |
| Packing Group           | II                                 |

## 15. Regulatory information

### International Inventories

| Component                                               | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Cerate(2-), hexakis(nitrato-O)-, diammonium, (OC-6-11)- | 16774-21-3 | X   | -    | X    | ACTIVE                                        | 240-827-6 | -      | -   |

| Component                                               | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Cerate(2-), hexakis(nitrato-O)-, diammonium, (OC-6-11)- | 16774-21-3 | X     | KE-09797 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

Authorisation/Restrictions according to EU REACH

Not applicable

Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                               | CAS-No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------------------------------|------------|----------|------------------------------|---------------------------|--------------------------------------------|
| Cerate(2-), hexakis(nitrato-O)-, diammonium, (OC-6-11)- | 16774-21-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                                               | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cerate(2-), hexakis(nitrato-O)-, diammonium, (OC-6-11)- | 16774-21-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
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www.thermofisher.com

**Creation Date**

24-November-2010

**Revision Date**

27-March-2024

**Print Date**

27-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**